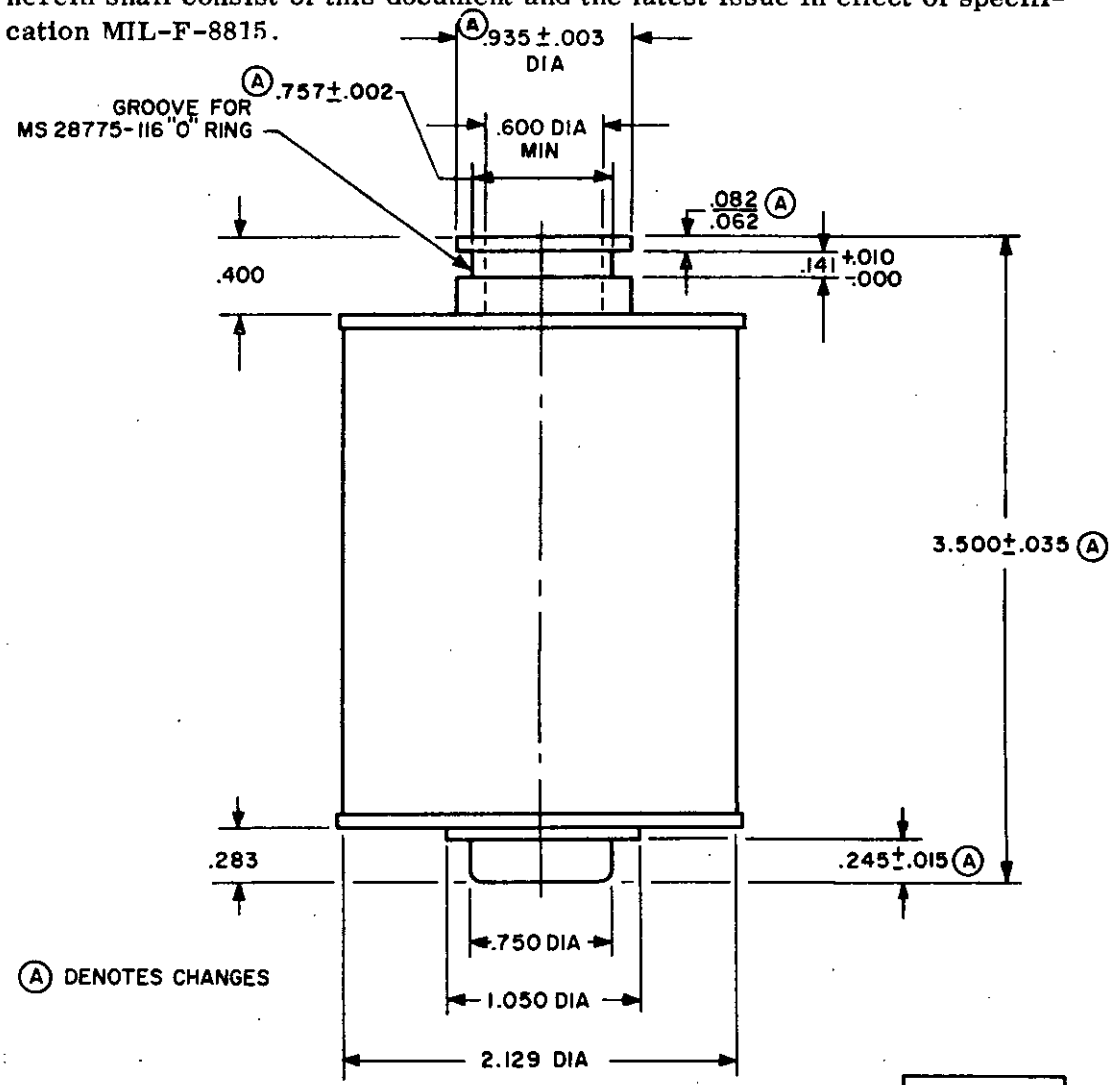
MIL-F-8815/21A(AS)
14 October 1976
SUPERSEDING
MIL-F-8815/21(AS)
30 July 1975

MILITARY SPECIFICATION SHEET

FILTER ELEMENTS, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 15 MICRON ABSOLUTE, AV8 AIRCRAFT, 10 GPM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of specification MIL-F-8815.



Ⓐ DENOTES CHANGES

FSC 1650

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NOTES: 1. Dimensions in inches, unless otherwise specified. See above Figure.

2. Tolerances, decimals ±.010, angles ±1°.

3. Unless otherwise specified, surface roughness shall not exceed 125 microinches, per MIL-STD-10. Remove all burrs and break sharp edges .005 minimum.

4. Material shall be in accordance with the procurement specification.

REQUIREMENTS:

Pressure Drop: Minimum pressure drop at rated flow across the filter element shall not exceed 5 psid.

Degree of Filtration: The filter element shall remove 100 percent of all contaminant larger than 15 microns and 98 percent of all test contaminant. The filter efficiency shall be a minimum of 98 percent.

Element Collapse: The filter element shall withstand a differential pressure of 150 psid.

Cold Start: Not applicable.

Maximum Particle Passed: The maximum particle passed shall be no greater than 15 microns.

Flow Fatigue: The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cpm maximum. (A)

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
100 cycles at 100 psid pressure drop (A)

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Part Number: The part number consists of the basic number of this specification sheet.

Example:

M8815/21

TABLE I. CONTAMINANTS

Flow Rate (GPM)	Degree of Filtration Add Rate	Maximum Particle Add Rate	Pressure Buildup		Migration Analysis Value
	F-9 Beads (gms)	F-9 Beads (gms)	Add Rate A C Dust (gms)	Minimum AC Dust to 18 psid (gms)	Maximum Allowable (gms)
10.0	1.20	0.05	1.0	7.0	0.007 Ⓐ

Intended Use: The filter element covered by this specification sheet is intended for use in the pump suction line of the model AV8 aircraft operating with hydraulic fluid conforming to MIL-H-5606.

Preparing activity:
Navy - AS

(Project No. 1650-N052-13)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-F-8815/21A(AS) FILTER ELEMENTS, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC, LINE, 15 MICRON ABSOLUTE, AV8 AIRCRAFT, 10 GPM

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
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